Week 3 Lab ENGN1218

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PRELAB QUESTIONS

$$
\begin{aligned}
15 - V_x &= (5\text{k } \Omega) I_1 \\
\frac{15 - V_x}{5\text{k } \Omega} &= I_1 \\
V_x &= (10\text{k } \Omega) I_2 \\
\frac{V_x}{10\text{k } \Omega} &= I_2 \\
V_x &= (4\text{k } \Omega) I_3 \\
\frac{V_x}{4\text{k } \Omega} &= 4 I_3 \\
I_1 - I_2 &= I_3 - 2 \text{ mA} \\
I_1 - I_2 &= I_3 - 2 \times 10^{-}3 \\
\frac{15 - V_x}{5\text{k } \Omega} - \frac{V_x}{10\text{k } \Omega} &= \frac{V_x}{4\text{k } \Omega} - 2 \times 10^{-}3 \\
V_x &\approx 9.09
\end{aligned}
$$

$$
\begin{aligned}
V_y &= V_{\text{ab}} \\
V &= IR \\
V_x - 10 &= I_1 \times 1 \\
V_x - 10 &= I_1 \\
V_x - V_y &= I_2 \times 1 \\
V_x - V_{\text{ab}} &= I_2 \\
V_y &= I_3 \times 1 \\
V_{\text{ab}} &= I_3 \\
KV_{\text{ab}} - I_1 &= I_2 \\
I_2 &= I_3 - 2\text{A}
\end{aligned}
$$

These are all our equations from V = IR, now to simultaneously solve

$$
\begin{aligned}
V_x - V_{\text{ab}} &= V_{\text{ab}} - 2 \\
V_x &= 2V_{\text{ab}} - 2 \\
KV_{\text{ab}} - 2V_{\text{ab}} - 2 + 10 &= 2V_{\text{ab}} - 2 - V_{\text{ab}} \\
V_{\text{ab}} &= 14 \\
V_{\text{ab}} &= V_y = V_z = 14 \\
V_x &= 2V_{\text{ab}} - 2 \\
V_x &= 2 \times 14 - 2 \\
V_x &= 26
\end{aligned}
$$

Now for current through resistors

$$\frac{V}{R} = I$$
$$\frac{V_x - 10}{R_1} = I_1$$
$$\frac{26 - 10}{1} = I_1$$
$$I_1 = 14$$
$$\frac{V_x - V_y}{R_2} = I_2$$
$$\frac{26 - 14}{1} = I_2$$
$$I_2 = 12$$
$$\frac{V_y}{R_3} = I_3$$
$$\frac{14}{1} = I_3$$
$$I_3 = 14$$

find thevenins resistance If you draw the circuit, it ends up being two 1 ohm resisters in series, so

$$1 + 1 = R$$
$$R = 2$$
$$\frac{V}{R} = I$$
$$\frac{14}{2} = I$$
$$I = 7$$